

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Electrical Engineering) SEMESTER - 5 Winter 2025 (Regular)

Course :Bachelor of Technology (Electrical Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 5

Subject Code & Name: BTEEPLE504A - HVDC

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
 2. Question No. 1 will be compulsory and include objective-type questions.
 3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6
 4. Use of non-programmable scientific calculators is allowed.
 5. Assume suitable data wherever necessary and mention it clearly.
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Q1. Objective type questions. (Compulsory Question) 12

- 1 The main advantage of HVDC transmission over HVAC transmission is
 - A. Higher charging current
 - B. Lower line losses for long distances
 - C. Requirement of reactive power
 - D. Higher corona loss
- 2 The main function of DC smoothing reactors is to
 - A. Control AC voltage
 - B. Reduce DC voltage
 - C. Reduce DC current ripple and faults
 - D. Improve power factor
- 3 Power reversal in an HVDC bridge converter is achieved by
 - A. Reversing DC voltage polarity
 - B. Reversing DC current direction
 - C. Changing firing angle control
 - D. Switching AC supply sequence
- 4 In actual control characteristics, the ideal control slope is modified mainly due to
 - A. Thyristor forward voltage drop
 - B. AC system harmonics
 - C. Commutation overlap and losses
 - D. DC smoothing reactor
- 5 A commutation failure in an HVDC converter mainly occurs due to
 - A. High DC voltage
 - B. Low AC voltage at inverter
 - C. High firing angle at rectifier
 - D. DC line open circuit

- 6 The main purpose of DC reactors in an HVDC system is to
 - A. Increase DC voltage
 - B. Limit DC fault current and reduce ripple
 - C. Improve AC side power factor
 - D. Reduce harmonics on AC side
- 7 The bypass valve in a converter bridge is triggered when
 - A. Power demand increases
 - B. Converter is in normal operation
 - C. There is a valve or commutation failure
 - D. DC voltage is stable
- 8 Harmonics that appear at predictable orders based on pulse number are called
 - A. Random harmonics
 - B. Interharmonics
 - C. Characteristic harmonics
 - D. Subharmonics
- 9 Offshore wind power evacuation prefers DC transmission mainly because
 - A. It uses simpler equipment
 - B. Cable charging current is eliminated
 - C. Transformers are unnecessary
 - D. Harmonics are absent
- 10 Hybrid transmission mainly combines
 - A. AC overhead lines and underground cables
 - B. Line-commutated and voltage-source converter technologies
 - C. Thermal and renewable sources
 - D. Series and shunt compensation
- 11 Power-flow analysis in an AC–DC interconnected system is mainly used to determine
 - A. Short-circuit current
 - B. Steady-state voltage and power distribution
 - C. Harmonic generation
 - D. Protection coordination
- 12 Application of DC-based links in wind power is preferred because
 - A. Wind speed remains constant
 - B. Power can be efficiently transferred over long distances
 - C. No converters are required
 - D. System requires less control

Q2. Solve the following.

- A) Explain Modern Trends in DC Transmission system 6
- B) Draw a neat and clean diagram of DC transmission system and mention all system components? Explain function of any four components. 6

Q3. Solve the following.

- A) Write the desired features of control and actual control characteristics? 6
- B) Write in detail about power reversal in HVDC transmission. 6

Q4. Solve Any Two of the following.

- A) Explain DC circuit breaker with the help of neat and clean diagram 6

- B) What are the typical causes and consequences of commutation failure in HVDC converter suggest mitigation strategies 6
- C) Write in brief on overvoltage oscillations 6
- Q5. Solve Any Two of the following.
- A) Differentiate between characteristic and uncharacteristic harmonics in HVDC systems. Provide example of each. 6
- B) Explain Introduction to Hybrid HVDC and off-shore .Wind power evacuation schemes. 6
- C) Write in detail about troubles due to harmonics? 6
- Q6. Solve Any Two of the following.
- A) Explain Power flow analysis of AC DC system. 6
- B) Discuss the methods used for transient and dynamic stability analysis in HVDC connected power system. How does HVDC influence system stability? 6
- C) Explain HVDC transmission application in wind power generation 6

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